

YAO-HUNG TSAI

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PROFILE

I am passionate and proactive, with proven expertise in **Numerical Simulation** and **Physics-based AI Models**. I specialize in refining computational methods to improve simulation accuracy and efficiency consistently. My goal is to apply these technologies to real-world engineering challenges, driving innovation and delivering effective, practical solutions.

EDUCATION

M.E. in Applied Mechanics | *National Taiwan University (NTU), Taiwan* Sep 2023 - June 2025 (Expected)

- Focus on Computational Fluid Dynamics (CFD), Physics-based AI Models, Parallel Computing

B.E. in Civil Engineering | *NTU, Taiwan* Sep 2019 - Jul 2023

- Focus on Fluid Mechanics, Numerical Simulations, Machine Learning & Deep Learning

EXPERIENCE

On - Campus | Research Assistant | *NTU, Taiwan* Sep 2023 - June 2025 (Expected)

- Designed and implemented CFD algorithms, developed physics-based AI models, and has published a Q1 paper.

Off - Campus | Hydraulic Engineering | *Stantec Taiwan Inc, Taiwan* Jul 2023 - Aug 2023

- Participated in engineering project bidding and designed a near-zero carbon emission project.

Off - Campus | On-Site Engineering | *New Asia Construction Corp, Taiwan* Jul 2022 - Aug 2022

- Participated in BIM building model inspections, interpreted CAD drawings, and managed on-site construction.

PROJECTS

Research Project | Physics-based AI Model | Data-Driven Turbulence Model Sep 2023 - Present

- Designed a physics-constrained AI model based on turbulence theory to ensure physically meaningful outputs.
- Leveraged AI's power to eliminate solving additional equations in traditional methods, enhancing efficiency.
- Integrated AI model with CFD solvers, reducing computational error by 78% compared to traditional methods.

Research Project | Numerical Simulations | Turbidity Current Dynamics Sep 2022 - Apr 2024

- Implemented a Fortran-based CFD (CPU-based) to simulate the interaction of 100 million particles with fluid.
- Improved efficiency and reduced runtime with Linked-list structures and MPI parallel computing.
- Investigated turbidity current speed, settling velocity, particle deposition, collision, and flow field development.

Research Project | Cluster Management | Cluster Administrator Sep 2022 - Present

- Managed an HPC cluster to facilitate large-scale numerical simulations and AI model training.
- Configured Linux systems for a secure, efficient setup, including virtualization, networking, and optimization.
- Resolved hardware and software issues to ensure stability and minimize downtime.

Side Project | Parallel Computing | High-Performance PDE Solvers Sep 2024 - Jan 2025

- Utilized CUDA parallel computing (GPU-based) to significantly enhance PDE computational efficiency.
- Integrated MPI techniques to optimize multi-node parallel computation, improving load balancing and scalability.
- Successfully solved three major PDEs: Wave Equation, Heat Equation, and Laplace Equation.

Side Project | Model integration | Fluid Simulation in NVIDIA Omniverse Jan 2025 - Present

- Simulated cylinder flows to generate datasets with varying Reynolds numbers and cylinder sizes.
- Trained a PINN surrogate model to predict flow fields for any Reynolds numbers and cylinder sizes.
- Integrated the PINN model into Omniverse for rapid visualization of flow fields.

TECHNICAL SKILLS

- Numerical Simulation** | C++ / CUDA / MPI / Fortran / MATLAB
- Machine Learning & Metaheuristics** | Python / PyTorch / TensorFlow / Scikit-Learn
- Operating Systems & Tools** | Linux / Windows / macOS / Git / Docker
- CFD & Engineering Software** | ANSYS Fluent / OpenFOAM / AutoCAD / SolidWorks / Omniverse

PUBLICATIONS

Tsai, Y.-H. & Chou, Y.-J. (2024). On the suspension and deposition within turbidity currents. *Journal of Fluid Mechanics*.

Tsai, Y.-H., Cheng, C.-J. & Chou, Y.-J. (Manuscript). Data-driven turbulence modeling by using tensor-based neural network.

ADDITIONAL ACTIVITIES

Oral Presentation | 48th Conference on Theoretical and Applied Mechanics | *Hsinchu, Taiwan* Nov 2024

Poster Presentation | 45th Ocean Engineering Conference | *Keelung, Taiwan* Nov 2023

Competition (4th Place) | Sustainable Campus Competition | *NTU, Taiwan* Jun 2022 - Dec 2022

GRANTS & FELLOWSHIPS

Graduate Scholarship | *NTU, M.E. in Applied Mechanics, Taiwan* 2023(First Place Admission), 2024

Research Assistant Fellowship | *National Science and Technology Council, Taiwan* 2023, 2024